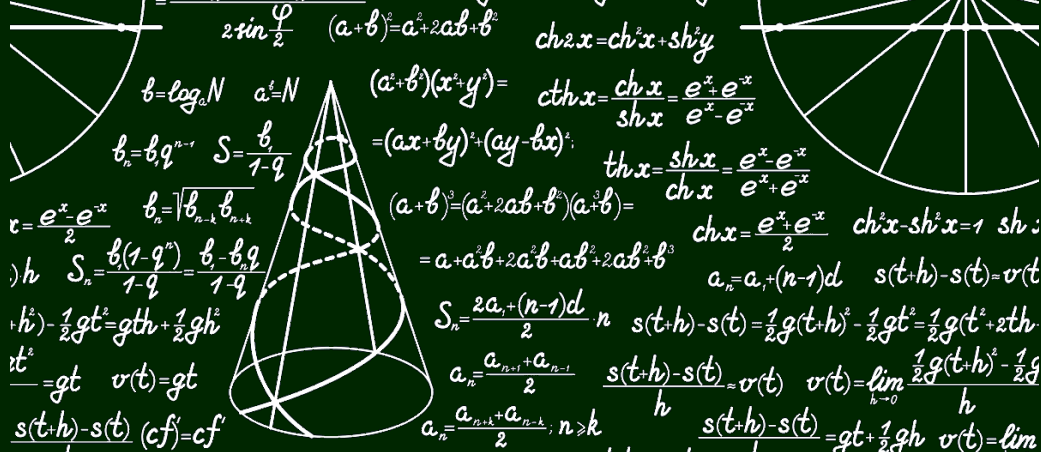




$$2 \sin \frac{\varphi}{2} \quad (a+b)^2 = a^2 + 2ab + b^2 \quad \cosh 2x = \cosh^2 x + \sinh^2 x$$

$$b = \log_a N \quad a^b = N \quad (a^2 + b^2)(x^2 + y^2) =$$

$$b_n = b_1 q^{n-1} \quad S = \frac{b_1}{1-q} \quad = (ax + by)^2 + (ay - bx)^2 \quad \cosh x = \frac{\cosh x + \sinh x}{e^x - e^{-x}}$$

$$r = \frac{e^x - e^{-x}}{2} \quad b_n = \sqrt{b_{n-k} b_{n+k}} \quad (a+b)^3 = (a^2 + 2ab + b^2)(a+b) =$$

$$h \quad S_n = \frac{b_1(1-q^n)}{1-q} = \frac{b_1 - b_n q}{1-q} \quad = a + a^2 b + 2a^2 b + ab^2 + 2ab^2 + b^3 \quad \cosh x = \frac{e^x + e^{-x}}{2} \quad \cosh^2 x - \sinh^2 x = 1 \quad \sinh :$$

$$h^2 - \frac{1}{2}gt^2 = gth + \frac{1}{2}gh^2 \quad S_n = \frac{2a_1 + (n-1)d}{2} \cdot n \quad s(t+h) - s(t) = \frac{1}{2}g(t+h)^2 - \frac{1}{2}gt^2 = \frac{1}{2}g(t^2 + 2th + h^2) - \frac{1}{2}gt^2$$

$$t^2 = gt \quad v(t) = gt \quad a_n = \frac{a_{n+1} + a_{n-1}}{2} \quad \frac{s(t+h) - s(t)}{h} \approx v(t) \quad v(t) = \lim_{h \rightarrow 0} \frac{\frac{1}{2}g(t+h)^2 - \frac{1}{2}gt^2}{h}$$

$$\frac{s(t+h) - s(t)}{h} (cf)' = cf' \quad a_n = \frac{a_{n+k} + a_{n-k}}{2}; n \geq k \quad \frac{s(t+h) - s(t)}{h} = gt + \frac{1}{2}gh \quad v(t) = \lim$$

MATH156 RC1

$$z = r(\cos \varphi + i \sin \varphi) = \cos \varphi + i \sin \varphi$$

$$a + bi \quad re^{i\varphi_1} re^{i\varphi_2} = r_1 r_2 e^{i(\varphi_1 + \varphi_2)} \quad \mathcal{M}(a, b)$$

$$a = r \cos \varphi \quad b = r \sin \varphi$$

$$s(t) = \frac{1}{2}gt^2$$

$$y = f(x_0) + f'(x_0)(x - x_0)$$

$$z = re^{i\varphi}$$

$$\arccos x = \frac{\pi}{2} - \arcsin x$$

Sincerest appreciation dedicated to 2024
VV156 TAs Haoran Shen, Zixuan Yang,
Chengyang Shi, Ruizhi Deng, as well as all
previous VV156 TAs.

1. Course Description

Topics of MATH156:

- Part1 Calculus I
 - Limits
 - Derivatives
 - Basic integrals
- Calculus II
 - Integration techniques
 - Infinite series

Future Honor Calculus:

- MATH255
 - Basic linear algebra
 - Partial derivatives & Multiple integrals
- MATH256
 - Differential equations
 - Deeper linear algebra
 - Fourier transform and Laplace transform

Other math courses in GC:

- MATH214
 - Linear algebra
- MATH203
 - Discrete math
- MATH401
 - Probabilistic methods
-

Professor: Runze Cai



Teaching Assistant:



Sophomore
ECE
Advanced
problems
beyond
lectures



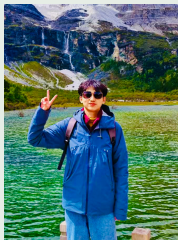
Sophomore
ECE
Problems
regarding
notes



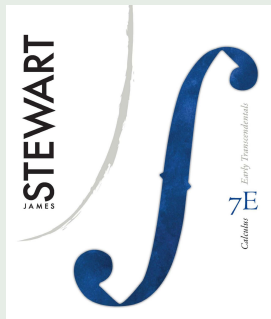
Sophomore
ECE
Homework
& In-class
problems



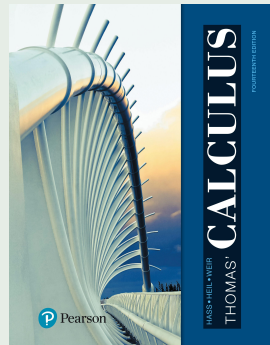
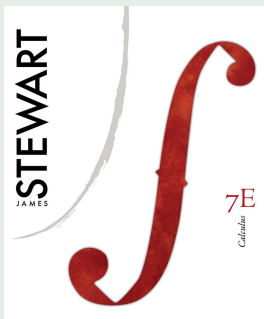
Sophomore
ECE
Problems
regarding
notes



Sophomore
ECE
Extra
exercises to
practice



Calculus by James Stewart
Official textbook



Thomas Calculus
A good Calculus book for
reference(recommended by
Hu Yuebo)

2. Sets

Definition

A set is a collection of different things; the things are elements or members of the set.

A set may be finite or infinite.

There is a unique set with no elements, called the empty set $\emptyset$.

A set with a single element is a singleton.

The commonly used number sets in mathematics include:

- $\mathbb{R}$: the set of all real numbers
- $\mathbb{N}$: the set of all natural numbers
- $\mathbb{Z}$: the set of all integers
- $\mathbb{Q}$: the set of all rational numbers
- $\mathbb{C}$: the set of all complex numbers

Definition

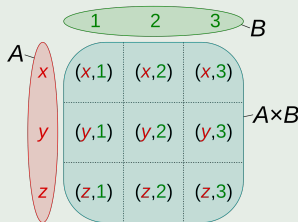
The Cartesian product of two sets $\mathbb{A}$ and $\mathbb{B}$, denoted $\mathbb{A} \times \mathbb{B}$, is the set of all ordered pairs (a, b) where a is an element of $\mathbb{A}$ and b is an element of $\mathbb{B}$. In terms of set-builder notation, that is

$$\mathbb{A} \times \mathbb{B} = \{(a, b) | a \in \mathbb{A} \text{ and } b \in \mathbb{B}\}$$

Using Cartesian products, $\mathbb{R}^2$ means

$$\mathbb{R} \times \mathbb{R} = \{(x, y) | x \in \mathbb{R} \text{ and } y \in \mathbb{R}\}$$

Geometrically, $\mathbb{R}^2$ represents the 2-dimensional Cartesian plane, where every point has coordinates (x, y) .



3. Functions

Definition

A **function** f is a rule that assigns to **each element** x in a set D exactly one element, called $f(x)$, in a set E .

Three essential factors of a function:

- Domain: D
A set containing the action objects of correspondence rule.
- Correspondence Rule
- Range: E
A set of all images corresponding to all elements in the domain under a certain correspondence rule.

The Vertical Line Test:

- A curve in the xy -plane is the graph of a function of x **iff** (if and only if) no vertical line intersects the curve more than once.

Four ways to represent a function

- Verbally (by a description in words)
- Numerically (by a table of values)
- Visually (by a graph)
- Algebraically (by an explicit formula)

Injection & Surjection & Bijection

- f is injective if $\forall x, y \in \mathbb{A}, x \neq y \rightarrow f(x) \neq f(y)$
- f is surjective if $\forall y \in \mathbb{B}, \exists x, f(x) = y$
- f is bijective if it is both injective & surjective

Symmetry

- Even function: $f(-x) = f(x)$
- Odd function: $f(-x) = -f(x)$

Tip: Give priority to whether the domain D of the function is symmetrical.

Increasing & Decreasing property

A function f is called **strictly increasing** on an interval I if

$$f(x_1) < f(x_2) \quad \text{whenever } x_1 < x_2 \text{ in } I$$

A function f is called **strictly decreasing** on an interval I if

$$f(x_1) > f(x_2) \quad \text{whenever } x_1 < x_2 \text{ in } I$$

- Monomial: $f(x) = x^n, i \in \mathbb{N}$
- Polynomial: $p(x) = \sum_{i=0}^n a_i x^i, n \in \mathbb{N}, a_i \in \mathbb{R}$
 n is the degree of the polynomial if $a_n \neq 0$.
- Power function: $f(x) = x^a, a \in \mathbb{R}$
Consider an arbitrary positive integer n :
 - $a = n, n > 0$: $y = x$: line $y = x^2$: parabola
 - $a = \frac{1}{n}, n > 0$: **root function**
 - $a = -1$: **reciprocal function**
- Rational function: $f(x) = \frac{P(x)}{Q(x)}, P, Q$ polynomials.
The domain consists of all values of x such that $Q(x) \neq 0$.
- Algebraic function: functions constructed using algebraic operations (such as addition, subtraction, multiplication, division, and taking roots) starting with polynomials **finitely** many times.

- Trigonometric function: $\sin, \cos, \tan, \cot(x) = \frac{1}{\tan(x)}, \sec = \frac{1}{\cos(x)}, \csc = \frac{1}{\sin(x)}$
 - $\sin^2(x) + \cos^2(x) = 1$
 - $1 + \tan^2(x) = \sec^2(x)$
 - $\cot^2(x) + 1 = \csc^2(x)$
- Exponential function: $f(x) = a^x, \quad a > 0$ In particular, $\exp(x) = e^x$
- Logarithmic function: $f(x) = \log_a x$, a is a positive constant and $a \neq 1$.
 $\log(x) = \exp^{-1}(x) = \log_e(x) = \ln x$
- Linear function
The graph of the function is a line $y = f(x) = mx + b$. m is the slope of the line and b is the y-intercept.
- Quadratic function
A polynomial of degree 2 is of the form $P(x) = ax^2 + bx + c$ and is called a **quadratic function**.

- Dirichlet Function $1_{\mathbb{Q}}(x) = \begin{cases} 1 & x \in \mathbb{Q} \\ 0 & x \notin \mathbb{Q} \end{cases}$

- Impulse Function $\delta(t) = \begin{cases} \infty & t = 0 \\ 0 & t \neq 0 \end{cases}$

- Step Function $H[n] = \begin{cases} 0, & n < 0 \\ 1, & n \geq 0 \end{cases}$

- Ramp Function $R(x) = \begin{cases} x, & x \geq 0 \\ 0, & x < 0 \end{cases}$

- Hyperbolic Function

- $\sinh(x) = \frac{e^x - e^{-x}}{2}$

- $\cosh(x) = \frac{e^x + e^{-x}}{2}$

- $\tanh(x) = \frac{\sinh(x)}{\cosh(x)}$

- Inverse trigonometric function

- $\operatorname{arsinh}(x) = \ln \left(x + \sqrt{x^2 + 1} \right)$

- $\operatorname{arcosh}(x) = \ln \left(x + \sqrt{x^2 - 1} \right)$

- $\operatorname{artanh}(x) = \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right)$

Vertical and Horizontal Shifts, suppose $c > 0$

$y = f(x) + c$, shift the graph of $y = f(x)$ a distance c units upward

$y = f(x) - c$, shift the graph of $y = f(x)$ a distance c units downward

$y = f(x - c)$, shift the graph of $y = f(x)$ a distance c units to the right

$y = f(x + c)$, shift the graph of $y = f(x)$ a distance c units to the left

Vertical and Horizontal Stretching and Reflecting, suppose $c > 1$

$y = cf(x)$, stretch the graph of $y = f(x)$ vertically by a factor of c

$y = (1/c)f(x)$, shrink the graph of $y = f(x)$ vertically by a factor of c

$y = f(cx)$, shrink the graph of $y = f(x)$ horizontally by a factor of c

$y = f(x/c)$, stretch the graph of $y = f(x)$ horizontally by a factor of c

$y = -f(x)$, reflect the graph of $y = f(x)$ about the x -axis

$y = f(-x)$, reflect the graph of $y = f(x)$ about the y -axis

Definition

The composition operator $\circ$ takes two functions, f and g , and returns a new function $h(x) := (g \circ f)(x) = g(f(x))$. Thus, the function g is applied after applying f to x . The expression $(g \circ f)$ is pronounced “the composition of g and f ”.

Composition of function is **associative**:

$$\forall x \in \mathbb{A}, h((g \circ f)x) = (h \circ g)f(x)$$

Composition of functions is **not necessarily commutative**.

In general, $f \circ g \neq g \circ f$

Question

- ① If $g(x) = 2x + 1$ and $h(x) = 4x^2 + 4x + 7$, find a function f such that $f \circ g = h$.
(Think about what operations you would have to perform on the formula for g to end up with the formula for h .)
- ② If $f(x) = 3x + 5$ and $h(x) = 3x^2 + 3x + 2$, find a function g such that $f \circ g = h$.

Solution:

$$\textcircled{1} \quad f(x) = x^2 + 6$$

$$\textcircled{2} \quad g(x) = x^2 + x - 1$$

Definition

Let f be a one-to-one function with domain A and range B . Then its **inverse function** f^{-1} has domain B and range A is defined by

$$f^{-1}(y) = x \Leftrightarrow f(x) = y$$

for any y in B .

- domain of $f^{-1} =$ range of f
- range of $f^{-1} =$ domain of f
- $f^{-1}(f(x)) \neq \frac{1}{f(x)}$

Question

Find a formula for the inverse of the function.

① $f(x) = 1 + \sqrt{2 + 3x}$

② $f(x) = \frac{4x - 1}{2x + 3}$

③ $f(x) = e^{2x-1}$

Solution:

$$\textcircled{1} \quad f^{-1} = \frac{1}{3}(x^2 - 1) - \frac{2}{3} \quad (x \geq 1)$$

$$\textcircled{2} \quad f^{-1}(x) = \frac{3x + 1}{4 - 2x}$$

$$\textcircled{3} \quad f^{-1}(x) = \frac{1}{2}(1 + \ln x)$$

- You can simply trace the dots for simple functions.
- You may also try to use software:
<https://www.geogebra.org>
<https://www.desmos.com>

4. Derivatives

Tangent Line

The **tangent line** to the curve $y = f(x)$ at the point $P(a, f(a))$ is the line through P with slope

$$m = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

provided that this limit exists.

Instantaneous rate of change

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

$$\text{instantaneous rate of change} = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \lim_{x_2 \rightarrow x_1} \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

Try to define velocity in this way.

Derivative

The **derivative of a function f at a number a** , denoted by $f'(a)$, is

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

if this limit exists.

The slope of the tangent line of a function is the corresponding derivative.

Common notations for derivatives:

- Newton: $\dot{y}$
- Leibniz: $\frac{dy}{dx}$ (at $x = a$: $\frac{dy}{dx}|_{x=a}$)
- Lagrange: $f'(x)$
- Jacobi: (Partial Derivatives) $\frac{\partial f}{\partial x}$

Differentiable

A function f is differentiable at a if $f'(a)$ exists. It is differentiable on an open interval (a, b) [or (a, ∞) or $(-\infty, a)$ or $(-\infty, \infty)$] if it is differentiable at every number in the interval.

Differentiable and Continuity

If f is differentiable at a , then f is continuous at a .

NOTE The converse of Theorem is false; that is, there are functions that are continuous but not differentiable. For instance, the function $f(x) = |x|$ is continuous at 0 because

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} |x| = 0 = f(0)$$

Linearity

$$\frac{d}{dx} (\alpha f(x) + \beta g(x)) = \alpha \frac{d}{dx} f(x) + \beta \frac{d}{dx} g(x)$$

Product Rule/ Leibniz Rule

$$\frac{d}{dx} [f(x)g(x)] = \left(\frac{d}{dx} f(x) \right) g(x) + f(x) \left(\frac{d}{dx} g(x) \right)$$

Quotient Rule

$$\frac{d}{dx} \left[\frac{f(x)}{g(x)} \right] = \frac{\left(\frac{d}{dx} f(x) \right) g(x) - f(x) \left(\frac{d}{dx} g(x) \right)}{g(x)^2}$$

$$\frac{d}{dx}(C) = 0$$

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

$$\frac{d}{dx}(cf(x)) = c \frac{d}{dx}f(x)$$

$$\frac{d}{dx}(\log_a x) = \frac{1}{x \ln a}$$

$$\frac{d}{dx}(\ln x) = \frac{1}{x}$$

$$\frac{d}{dx} \ln |x| = \frac{1}{x}$$

$$\frac{d}{dx}(\sin x) = \cos x$$

$$\frac{d}{dx}(\csc x) = -\csc x \cot x$$

$$\frac{d}{dx}(\cos x) = -\sin x$$

$$\frac{d}{dx}(\sec x) = \sec x \tan x$$

$$\frac{d}{dx}(\tan x) = \sec^2 x$$

$$\frac{d}{dx}(\cot x) = -\csc^2 x$$

$$\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}(\csc^{-1} x) = -\frac{1}{|x|\sqrt{x^2-1}}$$

$$\frac{d}{dx}(\cos^{-1} x) = -\frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}(\sec^{-1} x) = \frac{1}{|x|\sqrt{x^2-1}}$$

$$\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$$

$$\frac{d}{dx}(\cot^{-1} x) = -\frac{1}{1+x^2}$$

Question

Prove that $\frac{d}{dx} (\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$.

Solution: $y = \sin^{-1} x \Rightarrow \sin y = x$

$$\cos y \cdot \frac{dy}{dx} = 1 \quad (\text{Chain Rule})$$

$$\frac{dy}{dx} = \frac{1}{\cos y}$$

$$\cos^2 y = 1 - \sin^2 y = 1 - x^2$$

$$\cos y = \sqrt{1 - x^2}$$

$$\text{Therefore } \frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1 - x^2}}$$

Question

Find an equation of the tangent line to the curve $y = 2x \sin x$ at the point $(\frac{\pi}{2}, \pi)$.

Solution: $y' = 2(\sin x + x \cos x)$

$$k = y'\left(\frac{\pi}{2}\right) = 2$$

$$y - \pi = 2\left(x - \frac{\pi}{2}\right)$$

$$y - 2x = 0$$

Differentiate:

$$\textcircled{1} \quad y = x^3 + \frac{7}{x^4} - \frac{2}{x} + 12$$

$$\textcircled{2} \quad y = \sin x \cos x$$

$$\textcircled{3} \quad y = \sqrt{x} \sin x$$

$$\textcircled{4} \quad y = 3e^x \cos x$$

$$\textcircled{5} \quad y = \frac{x \sin x}{1 + x}$$

$$\textcircled{6} \quad y = \frac{1 - \sec x}{\tan x}$$

$$\textcircled{7} \quad y = x^2 \ln x \cos x$$

$$\textcircled{8} \quad y = \ln 3 + \frac{e^x}{x^2}$$

Solution:

$$\textcircled{1} 3x^2 - \frac{28}{x^5} + \frac{2}{x^2}$$

$$\textcircled{2} \cos 2x$$

$$\textcircled{3} \frac{1}{2}x^{-\frac{1}{2}} + x^{\frac{1}{2}}$$

$$\textcircled{4} 3e^x(-\sin x + \cos x)$$

$$\textcircled{5} \frac{dy}{dx} = \frac{(x \sin x)'(1+x) - x \sin x}{(1+x)^2} = \frac{\sin x + (1+x)x \cos x}{(1+x)^2}$$

$$\textcircled{6} y = \frac{\cos x - 1}{\sin x} \rightarrow \frac{dy}{dx} = -\frac{1}{1 + \cos x}$$

$$\textcircled{7} \frac{dy}{dx} = 2x \ln x \cos x + x \cos x - x^2 \ln x \sin x$$

$$\textcircled{8} (-2x^{-3} + x^{-2})e^x$$

Question

Let $y = \log_{\varphi(x)} f(x)$ ($\varphi(x) > 0$, $\varphi(x) \neq 1$, $f(x) > 0$). Suppose that both $\varphi(x)$ and $f(x)$ are differentiable. Calculate $\frac{dy}{dx}$.

Solution:

$$y = \frac{\ln f(x)}{\ln \varphi(x)}$$

$$\begin{aligned} \frac{dy}{dx} &= \frac{\frac{1}{f(x)} f'(x) \ln \varphi(x) - \frac{1}{\varphi(x)} \varphi'(x) \ln f(x)}{[\ln \varphi(x)]^2} \\ &= \frac{f'(x)}{f(x) \ln \varphi(x)} - \frac{\varphi'(x) \ln f(x)}{\varphi(x) [\ln \varphi(x)]^2} \end{aligned}$$

Question

Let $f(x) = xe^x$, prove that $f^{(n)}(x) = (x + n)e^x$ (taking derivatives n times)

Solution:

Pf. We want to prove that $\forall n \in \mathbb{N}, \quad f^{(n)}(x) = (x + n)e^x$

Let's prove it using induction on n .

Base case: $n = 0$, indeed

$$\text{LHS} = f^{(0)}(x) = f(x) = xe^x,$$

$$\text{RHS} = (x + 0)e^x = xe^x, \quad \text{so LHS} = \text{RHS}.$$

Inductive case: Assume the IH (inductive hypothesis): $f^{(n)}(x) = (x + n)e^x$

$$\begin{aligned} \text{Then } f^{(n+1)}(x) &= \frac{d}{dx} [f^{(n)}(x)] \stackrel{\text{IH}}{=} \frac{d}{dx} [(x + n)e^x] \\ &= 1 \cdot e^x + (x + n)e^x \\ &= e^x + (x + n)e^x \\ &= (x + (n + 1))e^x \end{aligned}$$

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$$\arccos x = \frac{\pi}{2} - \arcsin x$$